Controllers and Torque Allocators
AGH Racing DV Control

FS AGH Racing Driverless

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Contents

1 Scope and conventions

This document describes the shared implementation used by the packages:
- `dv_control`,
- `dv_skidpad_control`,
- `dv_acc_launch_control`.

All packages share the unbounded LTV MPC lateral controller. In the main package, the forward–backward (BF) profile generates the longitudinal reference.

The body-fixed convention is x forward, y left, and positive yaw counter-clockwise. Positive a_y is acceleration to the left and therefore loads the right-hand wheels. Positive a_x loads the rear axle. Wheel torque is expressed in N m; positive torque drives the vehicle.

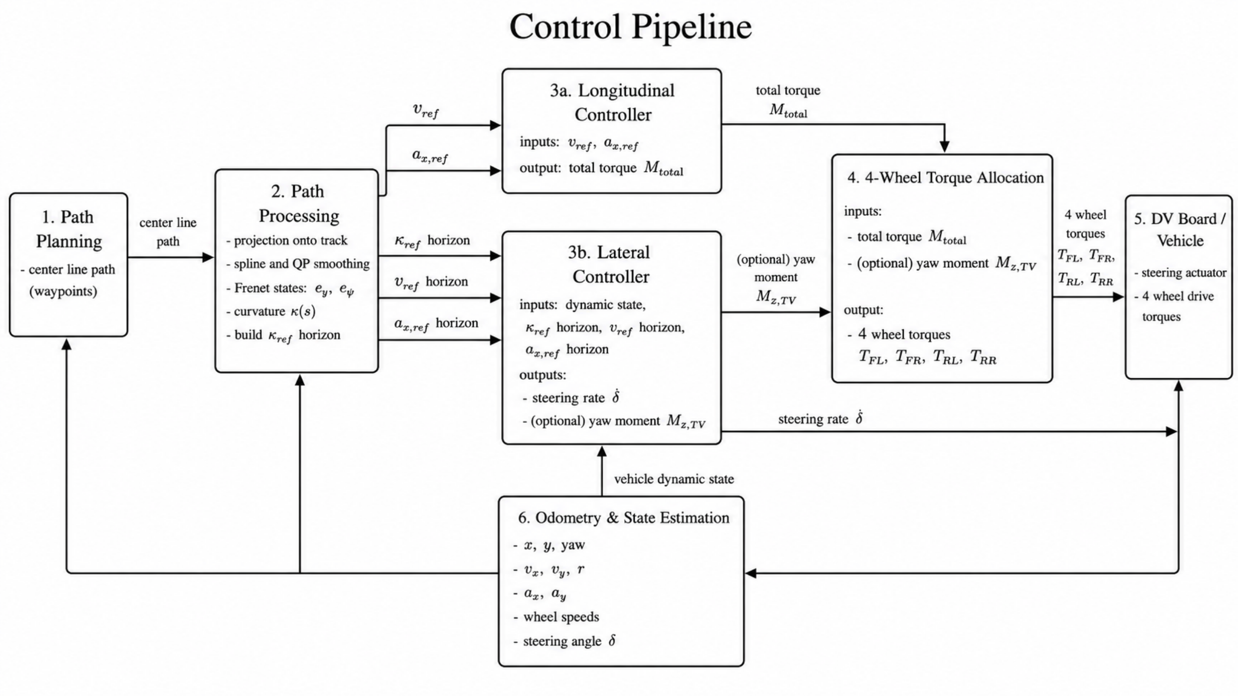

Figure 1: Control pipeline

Symbol	Unit	Meaning
e_y, e_ψ	m, rad	Frenet and heading errors
v_x, v_y, r	m/s, m/s, rad/s	body velocities and yaw rate
$\delta, \dot{\delta}, \delta_{\text{cmd}}$	rad, rad/s, rad	steering-system state
$M_{z,\text{TV}}$	N m	torque-vectoring yaw-moment request
l_f, l_r, L	m	CG-to-axle distances, $L = l_f + l_r$
t_f, t_r	m	front and rear track widths

2 Path processing and BF profile

The path is smoothed and parameterized by arc length s. Projecting the vehicle onto the spline gives e_y, e_ψ, and curvature $\kappa(s)$. The local speed limit imposed by lateral acceleration is

$$v_\kappa(s) = \min\left(v_{\max}, \sqrt{\frac{a_{y,\max}}{\max(|\kappa(s)|, \varepsilon)}}\right). \tag{1}$$

Lateral acceleration consumes part of the available tire friction. The longitudinal-acceleration limit is

$$a_y = v^2|\kappa|, \tag{2}$$

$$a_{x,\text{ell}} = a_{x,\max}\sqrt{\max\left(0, 1 - \left(\frac{a_y}{a_{y,\max}}\right)^2\right)}. \tag{3}$$

The power limit and resistance forces give

$$F_{\text{res}}(v) = C_d v^2 + C_{rr} m g, \tag{4}$$

$$a_{x,\text{drive}} = \min\left(a_{x,\text{ell}}, \max\left(0, \frac{P_{\text{drive}}/\max(|v|, v_\varepsilon) - F_{\text{res}}}{m}\right)\right), \tag{5}$$

$$a_{x,\text{brake}} = \min\left(a_{x,\text{ell}}, \max\left(0, \frac{P_{\text{brake}}/\max(|v|, v_\varepsilon) + F_{\text{res}}}{m}\right)\right). \tag{6}$$

For a spatial step Δs, the forward pass limits the reachable speed:

$$v_{i+1} \leftarrow \min\left(v_{i+1}, \sqrt{v_i^2 + 2a_{x,\text{drive,i}}\Delta s}\right), \tag{7}$$

and the backward pass guarantees that the vehicle can brake before a future constraint:

$$v_i \leftarrow \min\left(v_i, \sqrt{v_{i+1}^2 + 2a_{x,\text{brake,i}}\Delta s}\right). \tag{8}$$

The acceleration reference is computed without time differentiation:

$$a_{x,\text{ref,i}} = \frac{v_{i+1}^2 - v_i^2}{2\Delta s}. \tag{9}$$

The feedforward torque includes wheel rotational inertia:

$$m_{\text{eq}} = m + n_w\frac{I_w}{R_w^2}, \tag{10}$$

$$M_{\text{total,ff}} = m_{\text{eq}} a_{x,\text{ref}} R_w + M_{\text{res}}(v_x). \tag{11}$$

3 Shared unbounded LTV MPC

3.1 State and input

The optimizer state and input are

$$\boldsymbol{x} = \begin{bmatrix} e_y & e_\psi & v_y & r & \delta & \dot{\delta} & \delta_{\text{cmd}} & M_{z,\text{TV}} \end{bmatrix}^T, \tag{12}$$

$$\boldsymbol{u} = \begin{bmatrix} \dot{\delta}_{\text{cmd}} & \dot{M}_{z,\text{TV}} \end{bmatrix}^T. \tag{13}$$

No additional states are introduced between the command and the actuator model.

3.2 Nonlinear model used for linearization

The Frenet kinematics are

$$\dot{s} = \frac{v_x \cos e_\psi - v_y \sin e_\psi}{1 - \kappa e_y}, \tag{14}$$

$$\dot{e}_y = v_x \sin e_\psi + v_y \cos e_\psi, \tag{15}$$

$$\dot{e}_\psi = r - \kappa \dot{s}. \tag{16}$$

The velocity used in denominators is lower-bounded by $v_{\min}$ to avoid a numerical singularity.

The bicycle-model slip angles are

$$\alpha_f = \delta - \operatorname{atan2}(v_y + l_f r, v_x), \tag{17}$$

$$\alpha_r = -\operatorname{atan2}(v_y - l_r r, v_x). \tag{18}$$

The tire model is a simplified Magic Formula:

$$F_{y,f} = F_{z,f} D_f \sin(C_f \arctan(B_f \alpha_f)), \tag{19}$$

$$F_{y,r} = F_{z,r} D_r \sin(C_r \arctan(B_r \alpha_r)). \tag{20}$$

The lateral dynamics are

$$\dot{v}_y = \frac{F_{y,f} + F_{y,r}}{m} - v_x r, \tag{21}$$

$$\dot{r} = \frac{l_f F_{y,f} - l_r F_{y,r} + M_{z,\mathrm{TV}}}{I_z}. \tag{22}$$

The steering actuator is a second-order PT2 element. With $\nu_\delta \equiv \mathrm{d}\delta/\mathrm{d}t$:

$$\dot{\delta} = \nu_\delta, \tag{23}$$

$$\dot{\nu}_\delta = -\omega_s^2 \delta - 2\zeta_s \omega_s \nu_\delta + \omega_s^2 \delta_{\mathrm{cmd}}, \tag{24}$$

$$\dot{\delta}_{\mathrm{cmd}} = u_\delta, \qquad \dot{M}_{z,\mathrm{TV}} = u_M. \tag{25}$$

At the beginning of every iteration, δ is corrected using the physical encoder measurement; the encoder signal is not differentiated numerically.

3.3 Linearization, discretization, and QP

For successive nominal points along the horizon:

$$\begin{aligned} &\dot{\boldsymbol{x}} \approx A_c(\boldsymbol{x} - \bar{\boldsymbol{x}}) + B_c(\boldsymbol{u} - \bar{\boldsymbol{u}}) + f(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}}), \\ &A_c = \left.\frac{\partial f}{\partial x}\right|_{\bar{x},\bar{u}}, \qquad B_c = \left.\frac{\partial f}{\partial u}\right|_{\bar{x},\bar{u}}. \end{aligned} \tag{26}$$

Exact zero-order-hold discretization is

$$\exp\left(\begin{bmatrix} A_c & B_c \\ 0 & 0 \end{bmatrix} T_s\right) = \begin{bmatrix} A_d & B_d \\ 0 & I \end{bmatrix}. \tag{27}$$

The main package can optionally use the Tustin method.

The horizon cost contains state errors and command rates:

$$J = \frac{1}{2} \sum_{k=0}^{N-1} \left(\boldsymbol{x}_k^T Q \boldsymbol{x}_k + \boldsymbol{u}_k^T R \boldsymbol{u}_k\right) + \frac{1}{2} \boldsymbol{x}_N^T Q_N \boldsymbol{x}_N, \tag{28}$$

where Q weights e_y, e_ψ, v_y, r, while R weights $\dot{\delta}_{\text{cmd}}$ and $\dot{M}_{z,\text{TV}}$. The condensed problem has no inequality constraints:

$$\min_{\boldsymbol{U}} \ \frac{1}{2}\boldsymbol{U}^T H \boldsymbol{U} + \boldsymbol{g}^T \boldsymbol{U}, \qquad (H + \epsilon_H I)\boldsymbol{U} = -\boldsymbol{g}. \tag{29}$$

Angle and rate limits are applied to the first published action. The implementation uses preallocated buffers and matrix factorization without copying full matrices in the control loop.

3.4 Acceleration speed gate

In `dv_acc_launch_control`:

$$\delta_{\text{published}} = \begin{cases} 0, & v_x \le v_{\text{activation}}, \\ \text{integrate}\left(\dot{\delta}_{\text{cmd}}^{\text{MPC}}\right), & v_x > v_{\text{activation}}, \end{cases} \qquad v_{\text{activation}} = 2.0\,\text{m/s}. \tag{30}$$

The zero-output branch also resets the command state, the PT2 rate state, and the optimizer warm start. This prevents a stale command from causing a step after the activation threshold is crossed.

4 Wheel load transfer

4.1 Temporal relaxation

Raw acceleration measurements are not passed directly to the allocator. For $a \in \{a_x, a_y\}$, a first-order element is used:

$$\tau_{\text{load}}\dot{a}_f + a_f = a_{\text{measured}}. \tag{31}$$

The exact zero-order-hold update is

$$a_{f,k+1} = a_{f,k} + \left(1 - e^{-T_s/\tau_{\text{load}}}\right)(a_k - a_{f,k}). \tag{32}$$

The default is $\tau_{\text{load}} = 0.05\,\text{s}$, and the parameter is available in all three applications.

4.2 Static and longitudinal loads

With optional aerodynamic downforce:

$$F_{z,f}^{\text{axle}} = mg\frac{l_r}{L} + C_{l,f}v_x^2 - \Delta F_{z,x}, \tag{33}$$

$$F_{z,r}^{\text{axle}} = mg\frac{l_f}{L} + C_{l,r}v_x^2 + \Delta F_{z,x}, \tag{34}$$

$$\Delta F_{z,x} = \frac{m a_{x,f} h_{\text{CG}}}{L}. \tag{35}$$

4.3 Lateral transfer with roll centers and λ_ϕ

The roll-axis height at the CG longitudinal station is

$$h_{\text{RA}} = \frac{l_r h_{\text{RC,f}} + l_f h_{\text{RC,r}}}{L}. \tag{36}$$

The elastic roll moment is

$$M_{\phi,\text{el}} = m a_{y,f}(h_{\text{CG}} - h_{\text{RA}}). \tag{37}$$

The parameter $\lambda_\phi \in [0,1]$ is the fraction of the elastic roll moment assigned to the front axle. The transfers, defined as load moved from the inner wheel to the outer wheel, are

$$\Delta F_{z,y,f} = \frac{\left(m a_{y,f} \frac{l_r}{L}\right) h_{\mathrm{RC,f}} + \lambda_\phi M_{\phi,\mathrm{el}}}{t_f}, \tag{38}$$

$$\Delta F_{z,y,r} = \frac{\left(m a_{y,f} \frac{l_f}{L}\right) h_{\mathrm{RC,r}} + (1-\lambda_\phi) M_{\phi,\mathrm{el}}}{t_r}. \tag{39}$$

There is no additional factor of $1/2$. The wheel loads are

$$F_{z,\mathrm{FL}} = \frac{1}{2} F_{z,f}^{\mathrm{axle}} - \Delta F_{z,y,f}, \qquad F_{z,\mathrm{FR}} = \frac{1}{2} F_{z,f}^{\mathrm{axle}} + \Delta F_{z,y,f}, \tag{40}$$

$$F_{z,\mathrm{RL}} = \frac{1}{2} F_{z,r}^{\mathrm{axle}} - \Delta F_{z,y,r}, \qquad F_{z,\mathrm{RR}} = \frac{1}{2} F_{z,r}^{\mathrm{axle}} + \Delta F_{z,y,r}. \tag{41}$$

The transfer clamp preserves at least $F_{z,\mathrm{min}}$ on every wheel without changing the total load on either axle.

5 Torque allocation

5.1 QP allocator

The `DV_QP` mode is the default allocator in `dv_skidpad_control`. Its decision variable is the vector of wheel longitudinal forces

$$\mathbf{u} = \begin{bmatrix} F_{x,\mathrm{FL}} & F_{x,\mathrm{FR}} & F_{x,\mathrm{RL}} & F_{x,\mathrm{RR}} \end{bmatrix}^T. \tag{42}$$

The requested values form the vector

$$\mathbf{b} = \begin{bmatrix} F_x^\star & M_z^\star \end{bmatrix}^T, \qquad F_x^\star = \frac{M_{\mathrm{total}}}{R_w}. \tag{43}$$

The angles δ_L, δ_R are the physical front-wheel angles calculated from the bicycle steering angle using the fitted anti-Ackermann relation. The mapping from wheel forces to total body-frame longitudinal force and yaw moment is

$$\mathbf{G} = \begin{bmatrix} \cos\delta_L & \cos\delta_R & 1 & 1 \\ l_f \sin\delta_L - \frac{t}{2}\cos\delta_L & l_f \sin\delta_R + \frac{t}{2}\cos\delta_R & -\frac{t}{2} & \frac{t}{2} \end{bmatrix}. \tag{44}$$

Here, t is the common track width used by the allocator. The allocator minimizes the unconstrained quadratic objective

$$\min_{\mathbf{u}} \; (\mathbf{G}\mathbf{u} - \mathbf{b})^T \mathbf{Q} (\mathbf{G}\mathbf{u} - \mathbf{b}) + \mathbf{u}^T \mathbf{R} \mathbf{u}, \tag{45}$$

where

$$\mathbf{Q} = \mathrm{diag}\left(\frac{1}{F_{x,\mathrm{ref}}^2}, \frac{1}{M_{z,\mathrm{ref}}^2}\right), \tag{46}$$

$$\mathbf{R}_{ii} = \lambda_A \frac{\sum_j F_{z,j}}{\max(1\,\mathrm{N}, F_{z,i})} \frac{1}{F_{x,\mathrm{ref}}^2}. \tag{47}$$

The $\mathbf{R}$ weighting increases the force penalty on a lightly loaded wheel. The parameter λ_A regularizes the allocator and is distinct from λ_ϕ used by the lateral load-transfer model.

Expanding the objective gives

$$\mathbf{A} = \mathbf{G}^T\mathbf{Q}\mathbf{G} + \mathbf{R}, \qquad \mathbf{c} = \mathbf{G}^T\mathbf{Q}\mathbf{b}. \tag{48}$$

The stationarity condition gives

$$\mathbf{A}\mathbf{u}^\star = \mathbf{c}. \tag{49}$$

The system is solved using an LLT factorization. Before limiters, the wheel torques are $T_i = R_w u_i^\star$. A factorization or solution failure activates the axle-based fallback described below.

Inequality constraints are not part of this QP. After the solve, a single global scale factor limits the entire torque vector according to the friction ellipse, mechanical limits, total-torque rate limit, and final motor-torque limit. The common scale preserves the allocation-vector direction and the resulting distribution among wheels.

5.2 Base torque

In axle mode, and as the fallback after a QP failure, the total torque M is first subjected to direction, power, and rate limits. Half is assigned to each side of the vehicle and then distributed within each side in proportion to wheel load:

$$T^0_{\mathrm{FL}} = \frac{M}{2}\frac{F_{z,\mathrm{FL}}}{F_{z,\mathrm{FL}} + F_{z,\mathrm{RL}}}, \qquad T^0_{\mathrm{RL}} = \frac{M}{2}\frac{F_{z,\mathrm{RL}}}{F_{z,\mathrm{FL}} + F_{z,\mathrm{RL}}}, \tag{50}$$

$$T^0_{\mathrm{FR}} = \frac{M}{2}\frac{F_{z,\mathrm{FR}}}{F_{z,\mathrm{FR}} + F_{z,\mathrm{RR}}}, \qquad T^0_{\mathrm{RR}} = \frac{M}{2}\frac{F_{z,\mathrm{RR}}}{F_{z,\mathrm{FR}} + F_{z,\mathrm{RR}}}. \tag{51}$$

5.3 Yaw-moment realization

The load-transfer parameter λ_ϕ must not be confused with the auxiliary allocation scalar η_{TV}. The axle shares are

$$q_f = \frac{F_{z,\mathrm{FL}} + F_{z,\mathrm{FR}}}{\sum_i F_{z,i}}, \qquad q_r = 1 - q_f. \tag{52}$$

For wheel radius R_w, the yaw-moment coefficients are

$$C_f = \frac{l_f(\sin\delta_L - \sin\delta_R) - \frac{t_f}{2}(\cos\delta_L + \cos\delta_R)}{R_w}, \tag{53}$$

$$C_r = -\frac{t_r}{R_w}, \tag{54}$$

$$\eta_{\mathrm{TV}} = \frac{M_{z,\mathrm{TV}}}{C_f q_f + C_r q_r}. \tag{55}$$

The torque differences are

$$\Delta T_f = \eta_{\mathrm{TV}} q_f, \qquad \Delta T_r = \eta_{\mathrm{TV}} q_r, \tag{56}$$

$$T_{\mathrm{FL}} = T^0_{\mathrm{FL}} + \Delta T_f, \qquad T_{\mathrm{FR}} = T^0_{\mathrm{FR}} - \Delta T_f, \tag{57}$$

$$T_{\mathrm{RL}} = T^0_{\mathrm{RL}} + \Delta T_r, \qquad T_{\mathrm{RR}} = T^0_{\mathrm{RR}} - \Delta T_r. \tag{58}$$

Adding these differences does not change the total drive torque.

5.4 Friction ellipse and limits

For the estimated lateral force at each wheel:

$$|F_{x,i}|_{\max} = \mu_x F_{z,i} \sqrt{\max\left(0, 1 - \left(\frac{F_{y,i}}{\mu_y F_{z,i}}\right)^2\right)}, \qquad |T_i|_{\max} = R_w |F_{x,i}|_{\max}. \tag{59}$$

A single global scale factor $0 \leq \gamma \leq 1$ is selected so that all wheel torques satisfy the tire and mechanical limits simultaneously:

$$T_i^{\text{out}} = \gamma T_i. \tag{60}$$

The ellipse is disabled near standstill, where slip-angle estimation from v_x, v_y, r is ill-conditioned.

6 Critical parameters and tests

Key	Initial value	Validation
`model.mass_transfer.tau_load_s`	$0.05\,\mathrm{s}$	> 0
`model.body.h1_roll`	$0.08\,\mathrm{m}$	geometry measurement
`model.body.h2_roll`	$0.11\,\mathrm{m}$	geometry measurement
`model.body.lambda_phi_elastic_lateral`	0.479465	$[0, 1]$
`general.torque_allocation_lambda_factor`	0.075	> 0
`general.torque_allocation_fx_ref`	$1000\,\mathrm{N}$	> 0
`general.torque_allocation_mz_ref`	$10\,\mathrm{N\,m}$	> 0
`lateral_control.activation_speed_mps`	$2\,\mathrm{m/s}$	acceleration only

The unit test `common/test/load_transfer_test.cpp` checks the exact filter response after one τ, conservation of total normal force, conservation of axle-load sums during cornering, and increased right-wheel loads for positive a_y.